

Portfolio

Merna Aly

AI & Machine Learning Developer |
Frontend Developer

Introduction

WELCOME TO MY PORTFOLIO

AI-focused Computer Science student at the Faculty of Computer and Data Science, Alexandria University, with hands-on experience in React.js front-end development, Python programming, and Machine Learning. Experienced in building data-driven solutions, including an NLP-based career-matching application. Passionate about solving real-world problems through AI and software development, and eager to grow through practical projects and collaborative opportunities.



About Me

I'm Merna Aly, an AI-focused Computer Science student at the Faculty of Computer and Data Science, Alexandria University (GPA: 3.4). My technical skills include Python, Java, SQL, and front-end technologies such as React.js, JavaScript, HTML, and CSS.

Through NTI's Machine Learning & Data Analysis training, I gained hands-on experience in data preprocessing, model development, and evaluation, including building AI Career Navigator, an NLP-based career-matching application. I also have front-end development experience through DEPI training, where I worked on responsive web interfaces and UI/UX implementation.

I'm passionate about combining creativity with data-driven problem-solving to build practical solutions, and I'm eager to apply my skills, collaborate with others, and continue growing in AI and software development.



EDUCATION

Alexandria University
Faculty of Computers and Data
Science Department of Artificial Intelligence

2024-2028



My journey at Alexandria University's Artificial Intelligence department has been about more than just learning to code—it's been about learning how to think. Through Python, Java, and SQL courses, I've developed strong problem-solving skills and a deep appreciation for clean, logical code and efficient data structures.

This academic foundation has equipped me with a rigorous understanding of computer science principles, from algorithm analysis to database design. My AI specialization continues to shape my approach to development—whether building responsive front-end interfaces or working with complex data—ensuring my solutions are both technically sound and practically innovative.





Technical Skills

- Python
- Machine Learning
- Data Analysis
- Pandas
- Numpy
- Scikit-learn
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- React
- JavaScript
- SQL
- HTML5 &CSS3

Soft skills

- Problem Solving
- Communication
- Critical Thinking
- Teamwork
- Time Management

Languages

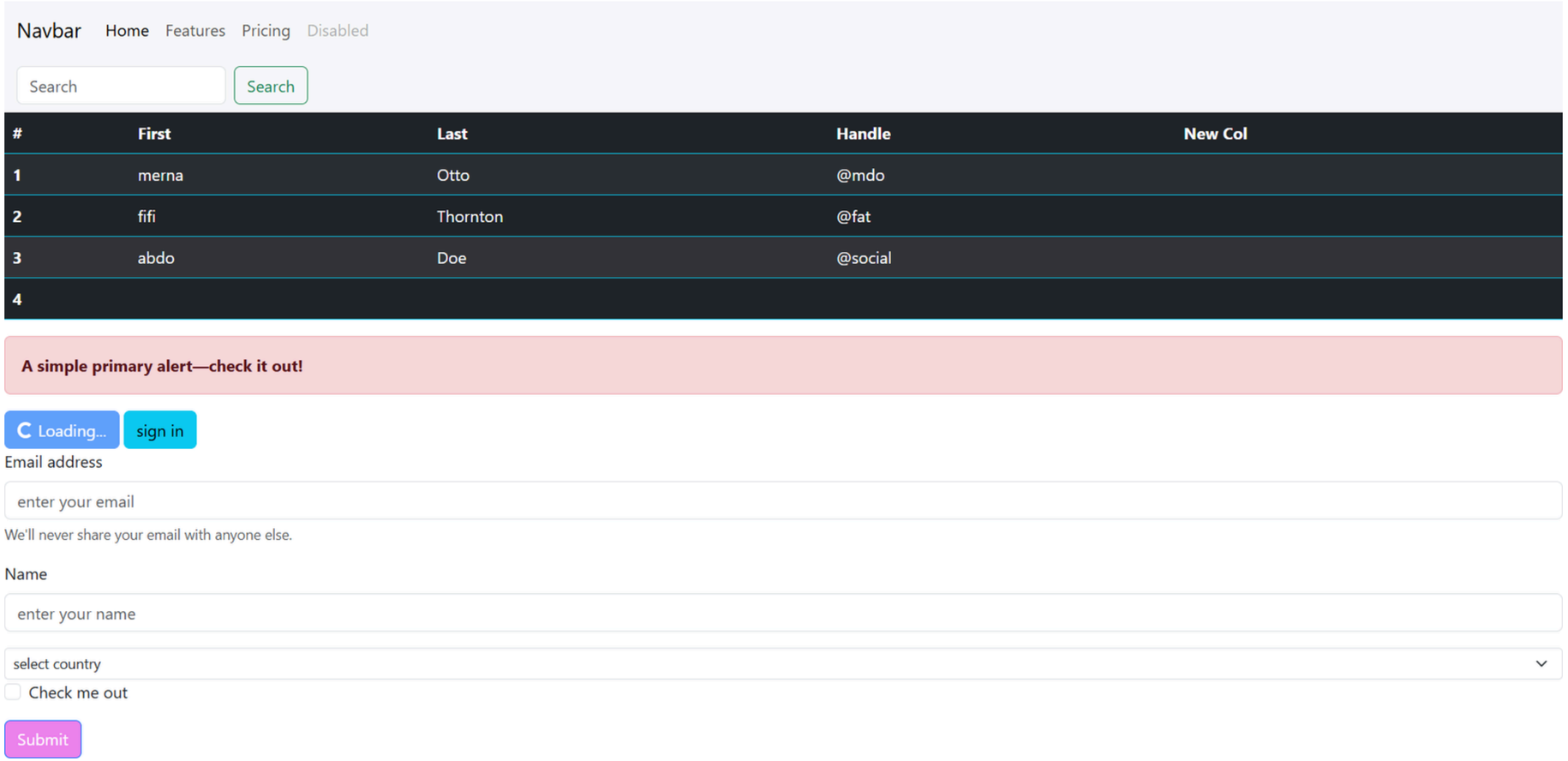
- English
- French



Projects

HTML, CSS, Bootstrap

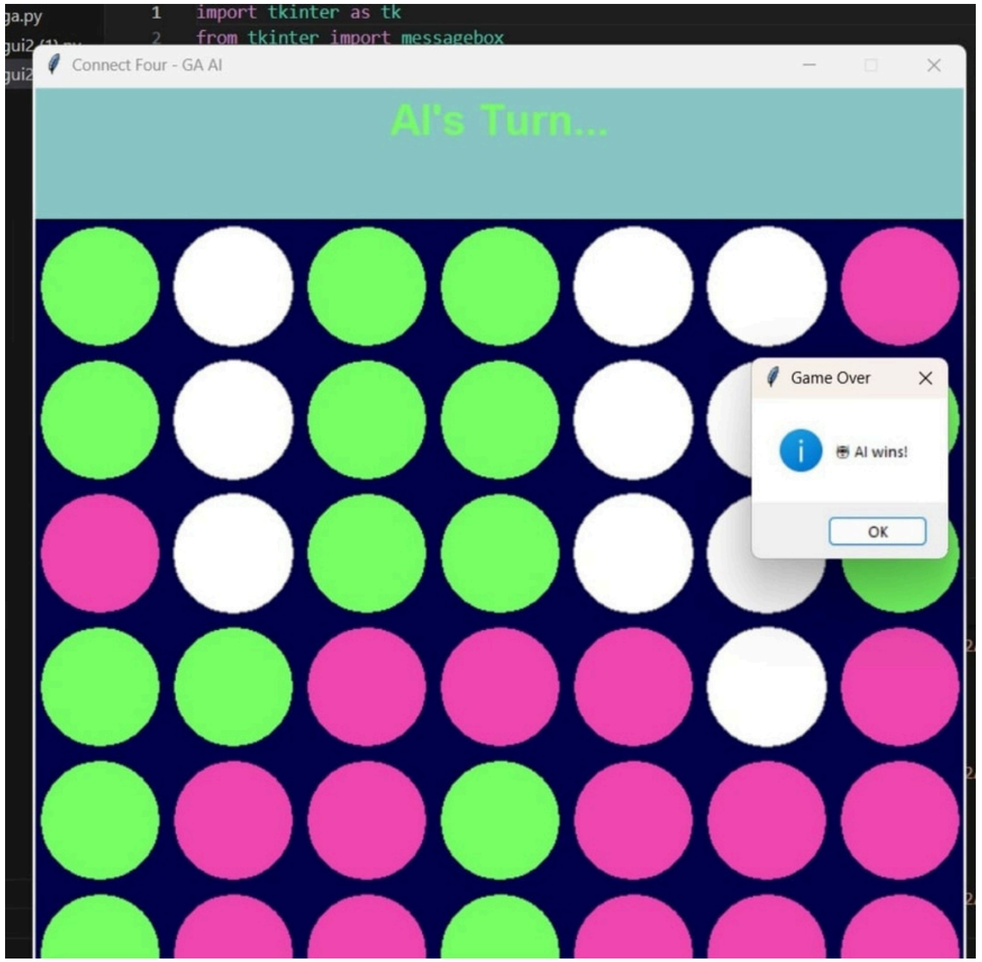
Built a simple responsive table layout to display user information. Includes basic form inputs and alert components for practice with Bootstrap's grid system and UI elements.

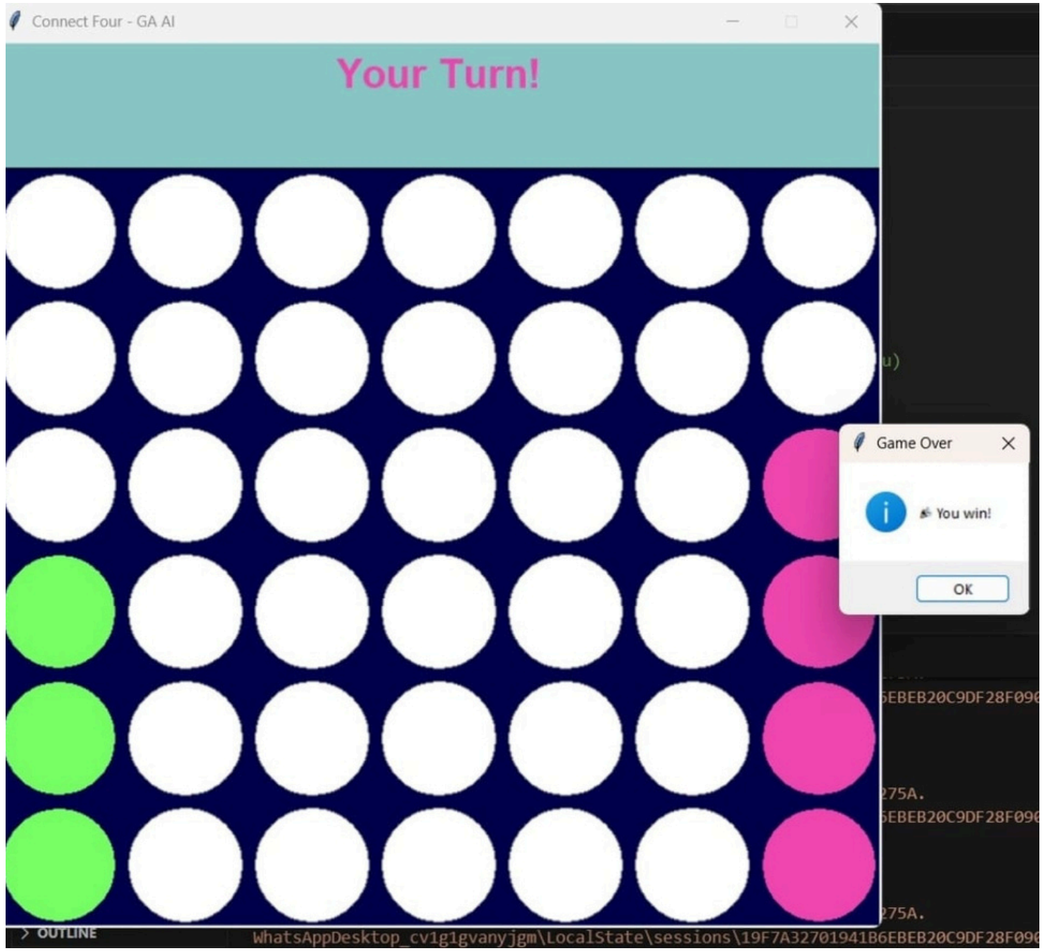


Smart Connect 4 - AI That Learns

Python, NumPy, Genetic Algorithms

Developed an AI-powered Connect 4 game that evolves and improves over time using a genetic algorithm. The AI learns optimal strategies by playing against itself, with each generation becoming smarter than the last.

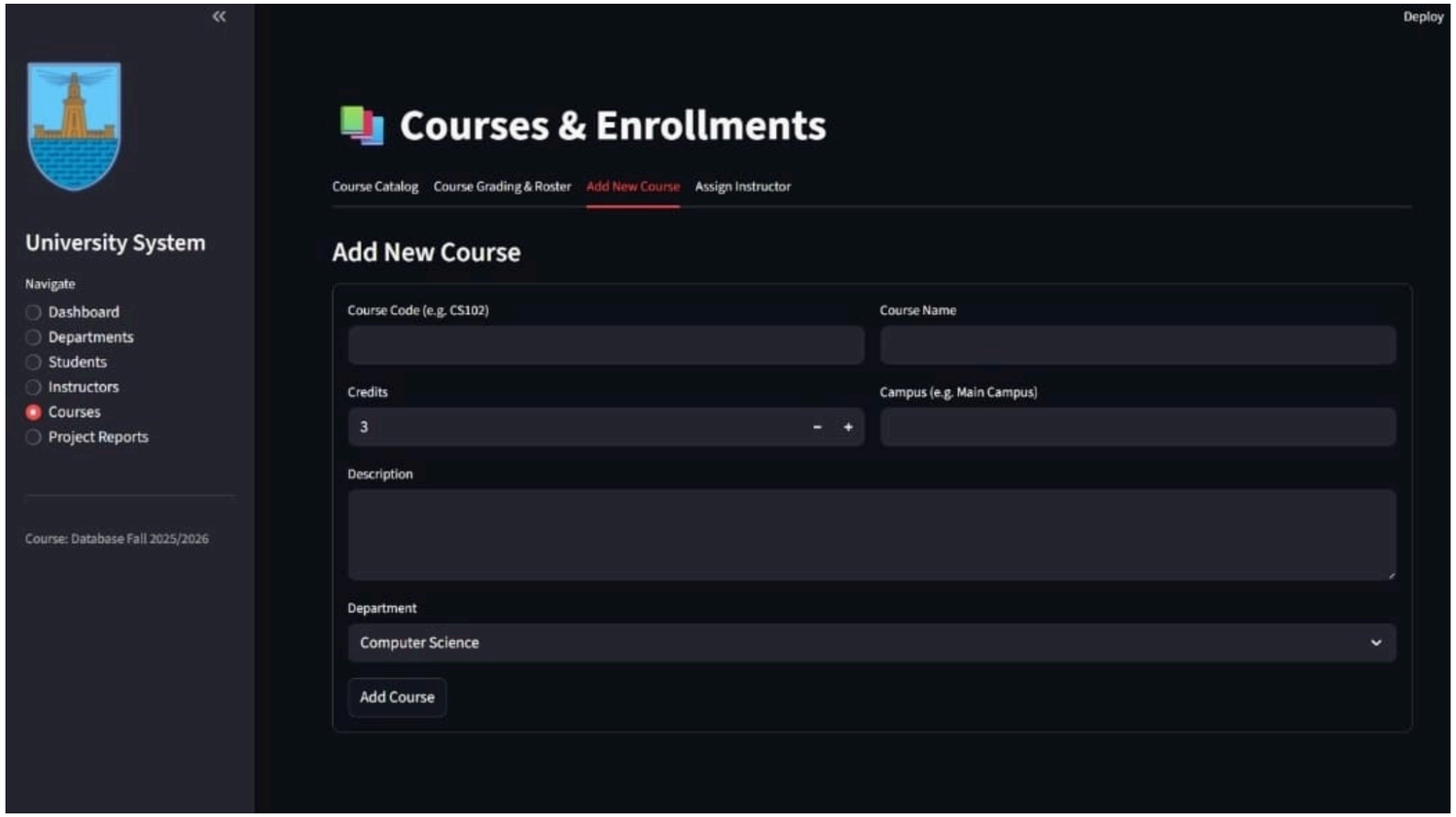


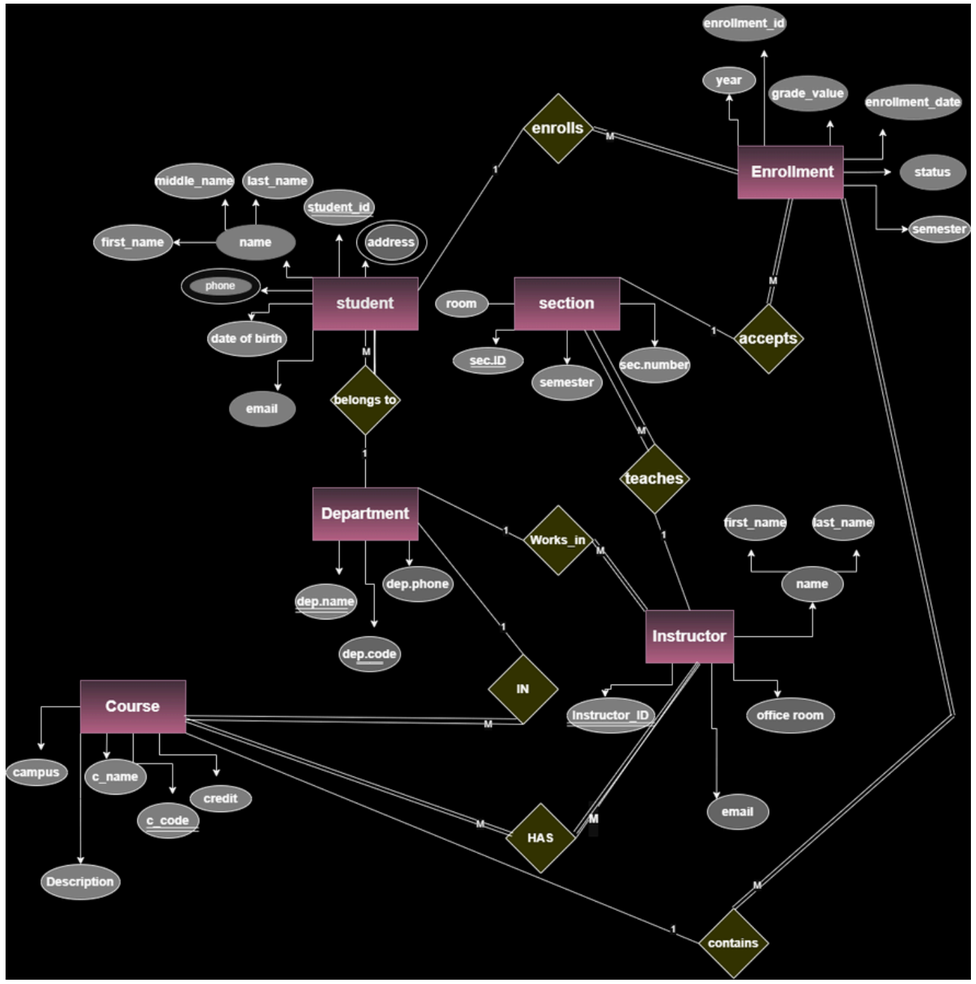


University Registration System

SQL, ER Diagram, Database Design

Designed and implemented a comprehensive **University Registration System** database to manage student enrollments, courses, instructors, and departments. The system handles the complete academic workflow—from student registration to grade tracking.





AI Career Navigator – Project Demo

This video demonstrates AI Career Navigator, a Machine Learning and Natural Language Processing (NLP) application developed as our final project for the NTI Machine Learning & Data Analysis training program. It showcases the application's main features, including CV analysis, skill extraction, matching candidates with relevant job opportunities using machine learning, and identifying missing skills to help users make informed career decisions.





Thank you for visiting, and I hope you find inspiration in my work!

Merna Aly

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